



AISAFETY: An AI-based smart system for enhancing operator safety in production processes

Francesco Di Paco^{a,*}, Luca Burattini^b, Roberto Gabbrielli^a, Luca Landi^b,
Francesco Marcelloni^c, Leonardo Marrazzini^a, Marco Palumbo^c, Marco Pirozzi^d

^a Department of Civil and Industrial Engineering, University of Pisa, Italy

^b Department of Engineering, University of Perugia, Italy

^c Department of Information Engineering, University of Pisa, Italy

^d Department of Technological Innovations and Safety of Plants, Products and Anthropic Settlements, INAIL, Italy

ARTICLE INFO

Keywords:

Artificial Intelligence
Computer Vision
RFID
Machinery Safety
Experimental setup

ABSTRACT

The forthcoming EU Machinery Regulation represents a significant evolution in industrial safety by enabling the use of Artificial Intelligence (AI) and autonomous technologies for safety-related functions in collaborative human-machine environments. However, these environments introduce new safety challenges related to human behaviour, improper machine use, and potential manumission of traditional safety measures.

The aim of this study is to propose an integrated AI-based supervision system, named AISAFETY, designed to enhance operator safety in production processes through continuous monitoring and intelligent decision support.

The proposed system is based on a layered architecture that integrates Radio-Frequency Identification for real-time operator localization, Computer Vision for visual monitoring, and a machine control layer governed by a rule-based expert system for managing safety-critical scenarios. The combination of heterogeneous sensing technologies provides redundancy and robustness in detecting hazardous conditions, such as unauthorized operator presence, tampering with safety guards, and improper collaborative robot operation. To support future validation, a prototype system has been implemented on an experimental setup consisting of a multimodal CNC lathe and a collaborative robot. A set of representative use cases has been defined to model potential hazardous situations and to guide the planned experimental evaluation of the system's safety supervision capabilities.

The proposed approach highlights the potential of AI- and IoT-based supervision systems to support the implementation of upcoming regulatory frameworks and to advance human-centred safety management in collaborative industrial environments.

1. Introduction

In January 2027, the new EU Machinery Regulation (MR) (European Parliament and of the Council, 2023) will come into effect, replacing the current Machinery Directive 2006/42/CE (MD) (European Parliament and of the Council, 2006). This updated legislation introduces notable changes to machine safety standards, including explicit provisions for the use of Artificial Intelligence (AI) in safety functions (see, for example, the essential health and safety requirement 1.2.1) and for the coexistence of humans and machines in collaborative environments – often referred to as collaborative robotics (see, for instance, the essential

health and safety requirement 1.3.7).

Traditionally, risk reduction strategies have relied on physical interventions, such as safety guards or barriers, while extraordinary situations (e.g., malfunctions or maintenance tasks) were handled using warning signals and operational instructions. However, the ongoing digital transformation (Industry 4.0) has spurred the development of advanced sensing and control systems, opening up new avenues for improving safety (Condry, 2023). In particular, the technical specification IEC TS 62998–1, “Safety of machinery – Safety-related sensors used for the protection of persons,” (IEC, 2019) provides guidelines for the design and integration of safety-related sensors (SRS) and sensor systems

* Corresponding author.

E-mail addresses: francesco.dipaco@phd.unipi.it (F.D. Paco), luca.burattini@dottorandi.unipg.it (L. Burattini), roberto.gabbrielli@unipi.it (R. Gabbrielli), luca.landi@unipg.it (L. Landi), francesco.marcelloni@unipi.it (F. Marcelloni), leonardo.marrazzini@unipi.it (L. Marrazzini), marco.palumbo@unipi.it (M. Palumbo), m.pirozzi@inail.it (M. Pirozzi).

<https://doi.org/10.1016/j.ssci.2026.107201>

Received 9 October 2025; Received in revised form 11 March 2026; Accepted 16 March 2026

Available online 20 March 2026

0925-7535/© 2026 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

(SRSS), emphasizing systematic capabilities in contexts not covered by specific regulatory standards. These innovative digital technologies have the potential not only to enhance safety but also to boost the efficiency of industrial operations (Cagno et al., 2024).

Among the most promising digital technologies in this domain, AI offers powerful tools for mitigating human error (Pishgar et al., 2021). It is now widely employed in risk assessment processes to leverage large datasets of accident records (Ayhan & Tokdemir, 2019; You et al., 2021). By automating the identification of hazardous scenarios and supporting data-driven decision-making, AI-based systems can substantially enhance operator safety, process reliability, and overall operational efficiency. According to Trivedi & Alqahtani (2024), the adoption of AI in occupational health and safety marks a crucial shift from reactive to proactive safety management, with the potential to significantly reduce the global burden of occupational injuries and fatalities in high-risk environments. This is particularly relevant for autonomous systems powered by AI, which can drastically limit human exposure to hazardous conditions (Buysse et al., 2025). Nevertheless, realizing these benefits and advancing risk management practices requires overcoming key challenges related to data quality, system explainability, cybersecurity, accountability gaps, and trust in AI models (Bostrom et al., 2024). Therefore, further research is needed to optimize the integration and governance of AI technologies within safety-critical applications (Stødle et al., 2025).

Within this context, the contribution of this work does not lie in proposing new technologies or novel AI models, but in the system-level integration of existing technologies within a unified AI-based safety control framework, explicitly designed to support autonomous safety functions in accordance with the emerging regulatory landscape. The proposed system combines heterogeneous sensing modalities, a knowledge-driven AI layer, and safety-related machine control into a single supervision architecture. Key features of the system include sensor redundancy to mitigate single-point failures, semantic reasoning through an ontology-based rule system to ensure transparency and explainability of safety decisions, and a closed-loop interaction with the machine control layer enabling timely and autonomous execution of safety functions.

The remainder of the paper is organized as follows. Section 2 presents the theoretical background of the study. Section 3 provides an overview of the system design both from hardware and software viewpoints. Section 4 outlines the operational scenarios considered for the testing phase, and Section 5 discusses the risk analysis approach adopted in the work. Finally, Section 6 concludes the paper and discusses the next steps of the on-going project.

2. Theoretical background

Recent research has increasingly explored AI-driven and sensor-based solutions to support occupational safety by detecting hazardous situations and unsafe behaviors in industrial environments. Early studies focused primarily on proximity-based risk mitigation using single sensing technologies. For example, Ruz et al. (2012) evaluated the safety performance of a Radio-Frequency Identification (RFID) prototype designed to prevent collisions between workers and robots under various hazardous scenarios. Similarly, Kim & Kim (2012) developed an RFID-based location-sensing system to support safety management of workers in the steel industry during crane lifting operations. Similarly, Reyes et al. (2014) proposed a wireless sensor-based monitoring system for machine guards in mining applications, aiming to reduce entanglement risks.

Subsequent works extended this paradigm through the adoption of wearable and IoT-based technologies. Choi et al. (2017) investigated the use of wearable sensors for monitoring construction workers' location and activity. Yang et al. (2020) introduced an IoT-based system for PPE-tool pairing to reduce hand-tool-related injuries. More recent contributions by Buffi et al. (2021) and Landi et al. (2021) employed

RFID-based positioning systems to detect the proximity of workers to potentially dangerous machinery. Rashidi et al. (2025) proposed smart PPE solutions integrating sensors into gloves and vests to monitor access to high-risk zones. Despite their effectiveness in controlled settings, these approaches typically rely on a single sensing modality and are strongly affected by environmental conditions, which limits their robustness and positional accuracy in complex industrial scenarios.

An emerging and rapidly expanding line of research in occupational safety focuses on the development of camera-based monitoring systems (Guo et al., 2021). Hwang et al. (2023) applied Computer Vision (CV) solutions to improve discriminative object localization algorithms, aiding safety managers with visualization to enhance indoor construction site safety management. Phatangare et al. (2023) implemented real-time human activity detection for multiple critical safety scenarios. Garyaev & Garyaev (2023) integrated AI with video surveillance systems to enhance efficiency and safety in the use of construction equipment. Similarly, Xu et al. (2023) proposed a CV-based system for monitoring safety harness usage on construction sites, targeting the prevention of falls from height, which remain the leading cause of accidents in this sector. More recently, Le & Lin (2025) developed an Augmented Reality-based alert system to detect and raise warning for near-miss incidents in construction sites with a 180-millisecond signal transmission (real-time monitoring). Barbosa et al. (2025) demonstrated the feasibility of identifying unsafe behaviors using CV-based monitoring of stair usage. Overall, the literature shows that most existing approaches emphasize object detection, tracking, or isolated activity recognition, typically addressing specific hazards or tasks. Far less attention has been devoted to the system-level integration of heterogeneous sensing technologies and to their coordinated use for autonomous safety supervision and control.

2.1. Research gaps

While previous studies demonstrate that AI, CV, RFID, and wearable technologies can individually contribute to improved occupational safety, several critical gaps remain. First, existing solutions are largely technology-centric and modular, addressing isolated safety functions rather than adopting a holistic, integrated perspective on safety supervision. Second, the combined and concurrent use of complementary sensing technologies has not yet been systematically explored to enhance robustness and reduce false negatives in safety-critical decisions. Third, most systems focus on monitoring and alerting, with limited capability for autonomous intervention through direct interaction with machine control systems.

To address this gap, the present study introduces an innovative integrated system for the autonomous control and monitoring of industrial working environments, leveraging AI to enhance operational safety in production processes. In particular, the proposed system provides:

- A multilayer supervision architecture that combines CV and RFID data to improve the reliability of safety monitoring in industrial environments;
- An AI module based on a rule-based expert system and ontology, used to formally represent safety knowledge, regulatory constraints, and operational rules;
- A closed-loop connection between risk assessment and machine control, allowing the system to autonomously trigger safety-related actions rather than only issuing warnings.

To the Authors' knowledge, this is the first integrated safety supervision system explicitly conceived to comply with the requirements of the forthcoming EU MR, particularly with respect to AI-enabled safety functions and human-machine collaboration. As a result, the paper contributes a methodological and architectural framework for developing regulation-aware, deployable AI-based safety systems, advancing beyond isolated or single-technology solutions commonly reported in

the literature.

3. Description of the system

The production system under analysis consists of a multimodal CNC lathe and a robot, which collaborate to handle and machine workpieces. The robot performs a pick-and-place operation to position the workpiece next to the operator. The operator then places the piece into the lathe chuck, after which the cobot closes the lathe door. The machining phase is initiated via an electrical signal exchanged between the control cabinet, the cobot, and the lathe. In compliance with EN ISO 12100 (ISO, 2010), the lathe is equipped with interlocking guards featuring a start function, ensuring that it can only begin operation without direct operator consent under specific safety conditions. Upon completing the machining process, the workpiece is manually removed and subsequently stored by the cobot in a designated location based on the process outcome.

As pertains to the CNC lathe, it is equipped with a PLC safety module, which effectively manages all safety Inputs and Outputs (I/Os). The lathe includes four distinct fairings, as illustrated in Fig. 1. The guards for the door and spindle are movable (Fig. 1(a)) and are therefore equipped with switches designed to detect both the status of the fairings and their locking mechanisms. These systems operate in accordance with the machine's operational mode through the integration of interlocks and mechanical blocks. In contrast, the carriage and side guards are fixed (Fig. 1(b)). The lathe is designed to operate under four distinct modes, as specified in the machine's user manual:

- **Manual Mode:** Allows manual processing with the door guard open while other guards remain in place. Requires a simple lathe operator, experienced operator, or tool technician.
- **Automatic Mode:** Enables automatic CNC processing with the spindle guard open while other guards remain in place. Inexperienced operators may perform part replacement tasks only when the machine is stopped.
- **Setup Mode:** Used for machine preparation, with the carriage guard in place but other guards open or removed as needed. Involves a simple lathe operator, experienced operator, tool technician, or maintenance operator.
- **Service Mode:** Designed for cleaning and lubrication under safe conditions, with no guards required. Only maintenance operators are authorized to use this mode.

These operational modes are essential for identifying potential hazards and conducting a comprehensive risk assessment. As far as the robot is concerned, it is a UR3e collaborative robot (cobot). It consists of six cylindrical joints, providing six degrees of freedom (Fig. 2).

Each joint is equipped with an electric motor with a rotational range between -360° and $+360^\circ$. The cobot is linked to the working table through a base and to a changeable end-effector via a tool flange,

offering significant flexibility in terms of operations that can be performed Fig. 3 illustrates the layout of the machines and the placement of walls within the hypothetical workroom. This initial configuration is essential for defining the workspace, which is divided into three distinct zones based on the proximity of the operator to the equipment:

- **Zone 1:** The area is directly adjacent to the lathe.
- **Zone 2:** The region encompassing the workpiece holders, both pre- and post-turning, as well as the circular area traversed by the robot during its movements.
- **Zone 3:** The space where the operator interacts with and controls the machine assembly via the command console.

This zoning approach provides a clear framework for understanding the interactions between the operator, the lathe, and the robot within the defined workspace.

The risk assessment for the described workspace is performed using the well-established “bottom-up” approach outlined in the ISO 12100 standard (ISO, 2012). This method begins by defining all possible operational states of the system, followed by the identification of hazard and associated risks for each state. For the activity of the robot, two operational states are considered: full speed and reduced speed. However, as the consequences of impacts at reduced speed have not yet been analyzed, the robot is categorized for this assessment as either stationary or moving. During the design phase, criteria must be established to determine under which conditions movement is enabled or disabled. To address this requirement, an enable signal has been integrated into the system to control the robot's motion.

While many risks are mitigated through traditional safety functions, in compliance with MD and standards such as ISO 13849 (ISO, 2015), some hazards remain unresolved by the current state of the art. These unresolved hazards, relevant to the system under study, are presented as “Use Cases” (UC) in Section 4. Typically, these scenarios are classified as residual risks, with mitigation limited to providing warnings or information to the operator. However, the introduction of the new EU MR creates opportunities to apply innovative technologies for advanced risk reduction. The next section details the proposed monitoring and supervision system and its key components, specifically designed to address these challenges and deliver enhanced safety performance.

3.1. IoT Sensors Description

This subsection introduces the IoT sensors integrated into the proposed system and their roles within the experimental monitoring and supervision setup. The preliminary design incorporates the following key elements:

- **Wearable RFID Devices:** Objects and Operators are equipped with RFID tags. These tags operate in the UHF band, allowing long-range communication with the RFID antennas. The tags are passive,

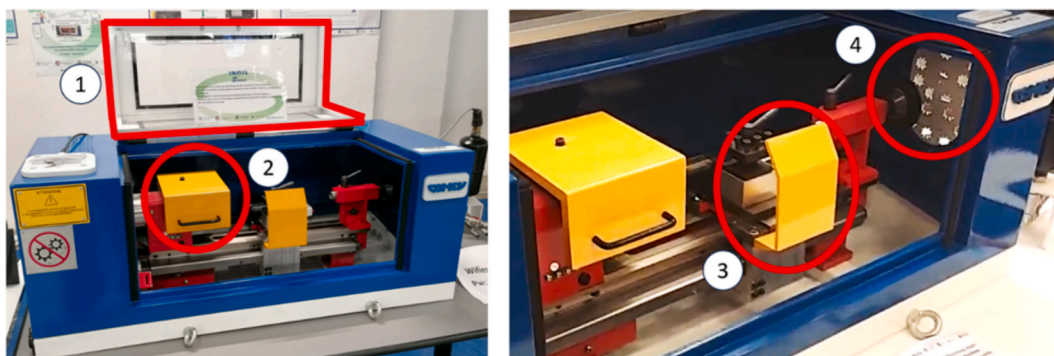


Fig. 1. (a) Door guard (1) and spindle guard (2); (b) carriage guard (3) and side guard (4).

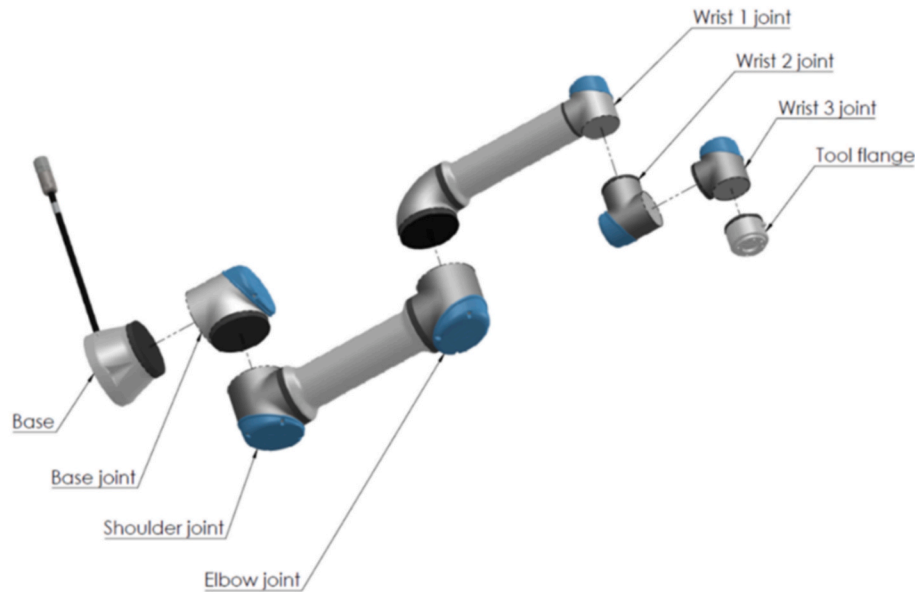


Fig. 2. Exploded view of UR robotic arm.

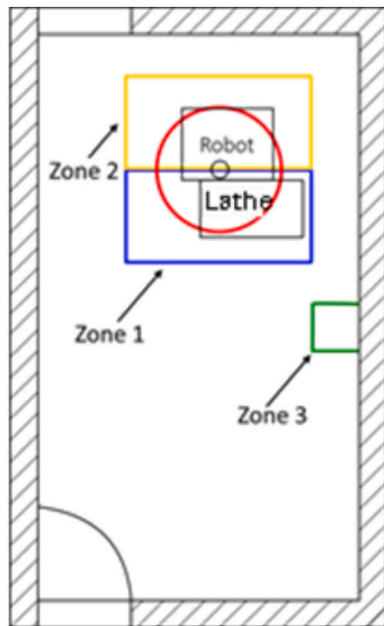


Fig. 3. Workspace surrounding the prototype.

meaning they do not require a battery; instead, they harvest power from the electromagnetic signal emitted by the antennas. Each tag has a unique identifier, enabling precise tracking of individual operators and objects. A network of six strategically positioned RFID antennas is installed around the machine and within the workspace. These antennas are responsible for emitting signals that activate the tags and for receiving the signals reflected back by the tags. The system uses the received signal strength indicator (RSSI) and phase measurements to estimate the position of the tags within the workspace. This information is processed to provide real-time location tracking of operators and objects, ensuring accurate monitoring of their positions and movements.

- CV system: Cameras continuously monitor the work areas, utilizing advanced image recognition algorithms to identify potential risks to operators. This system employs visible-spectrum cameras and

advanced microcomputers, specifically Raspberry Pi 5 devices, to identify potentially hazardous anomalies. These devices feature high-definition cameras and utilize strategically placed ArUco Markers to define work areas and enhance detection accuracy (Fig. 4). ArUco Markers, encoded black-and-white patterns similar to QR codes, are leveraged for their simplicity, cost-effectiveness, and robustness in real-time applications. Their advantages include rapid identification, low cost, flexibility in size and material, and reliable performance even under challenging conditions such as partial occlusion or unfavorable angles. Data communication between the Raspberry Pi devices and the AI module occurs via Wi-Fi using REST HTTP protocols. Each Raspberry Pi operates independently, continuously collecting and analyzing data, which is synchronized and sent to the AI module in real-time. The CV system uses YOLOv8 (You Only Look Once, version 8), a highly accurate and fast object detection model implemented on Raspberry Pi 5 devices with 8GB of RAM. The model, considering data acquisition, elaboration and decision on the actual operative decision, processes images at approximately 200 milliseconds per frame, enabling real-time monitoring of the work area. Taking the international standard EN ISO 13855:2024 “Safety of machinery. Positioning of safeguards with respect to the approach of the human body” as reference, the separation distance can be consequently estimated equal to about 400 mm and 320 mm for the hand speed value and walking speed value respectively, which are compatible with the dimension of the pilot machine. So, the processing time appears suitable for assuring effective protection for the operator.

ArUco Markers are also used to delineate the boundaries of operational zones, with markers positioned at the corners of these areas. The combined use of YOLOv8 and ArUco Markers ensures the CV system maintains high safety standards while enhancing situational awareness in the work environment. The CV system not only precisely locates machinery and personnel but also monitors and flags abnormal behaviors or risks, enabling predefined safety measures to improve worker safety and operational efficiency.

- Safety-Related Software: Data collected from wearable devices and cameras is processed by safety software, which allows supervisory operators to monitor system functionality (Fig. 5). In compliance with point 1.2.1, Annex III of MR, the software can intervene to

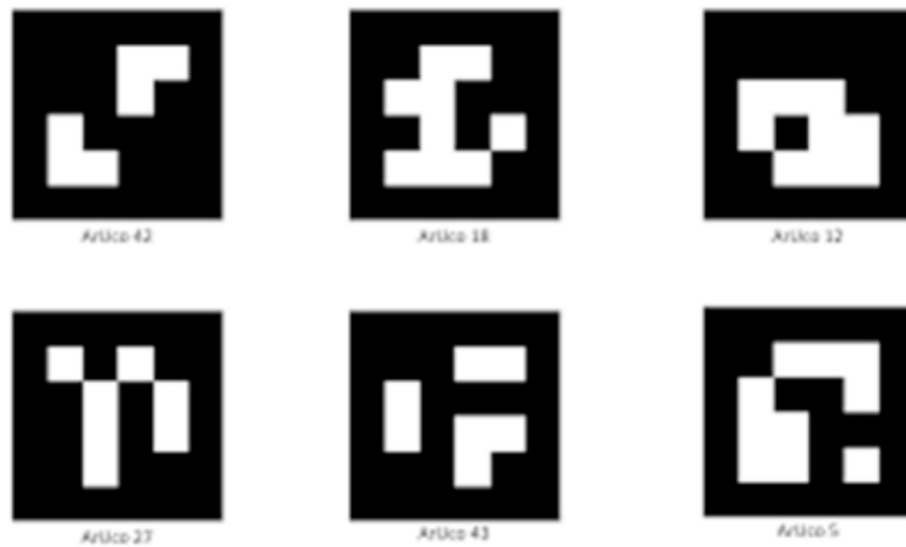


Fig. 4. Examples of ArUco markers.

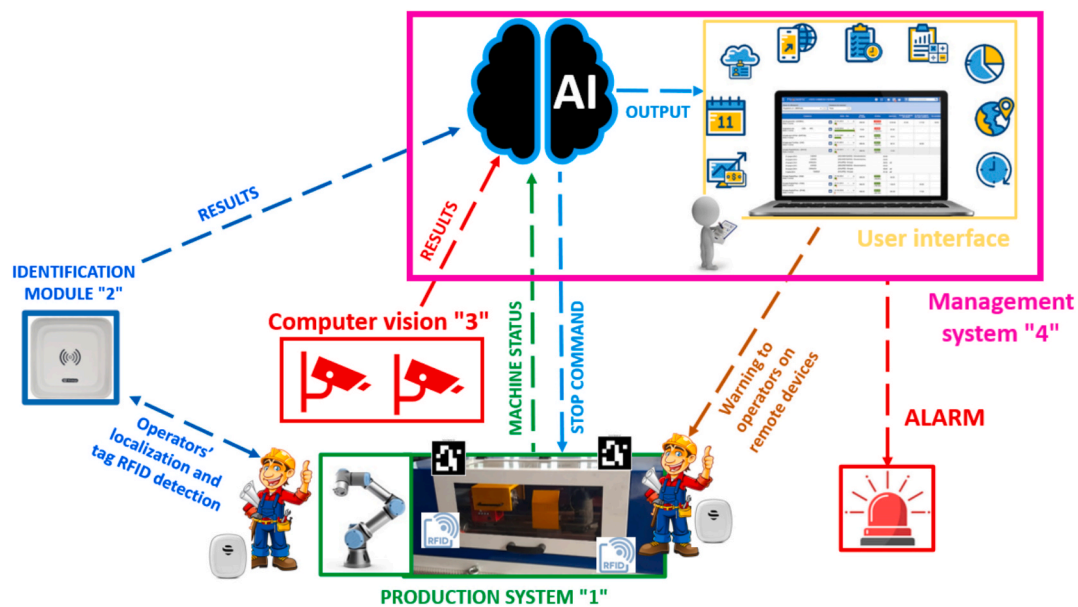


Fig. 5. Prototype scheme components and their relationship.

ensure safety, particularly during emergencies, by initiating corrective actions.

To explore the benefits of these advanced technologies in the workplace, a preliminary system design has been developed. Based on earlier considerations, this framework emphasizes the critical role of integrating innovative technologies. Autonomous devices equipped with AI modules and computational capabilities play a key role in processing data and calculating the Performance Level (PL) of the safety functions.

Although determining the PL of the monitoring and supervision system poses challenges, segregating sensor outputs ensures a robust foundation for analysis. Data from each sensor is independently analyzed before being transmitted to the AI module, which is responsible for decision-making. This integration of IoT sensors and AI systems enables the machine to interpret new operational scenarios, detect potential risks to operators, and execute commands, such as halting machine operations when necessary. The interactions between system components are illustrated in Fig. 5. The integrated system, combining

RFID, computer vision (CV), and AI technologies, is capable of identifying hazardous conditions—such as faults, malfunctions, or improper actions—based on the machine's operational status. Upon detecting risks, the monitoring and supervision system communicates with the machine control system to ensure safety of the operator, for instance, by triggering an emergency shutdown. This approach underscores the transformative potential of IoT and AI in creating smarter, safer, and more responsive work environments.

3.2. Description of the AI module

The AI module was developed as a Rule-Based Expert System (Tan, 2017) to store knowledge acquired from domain experts and the MR and to make decisions when specific conditions are met. The architecture of the Expert System consists of three primary components:

- Knowledge Base (KB): The KB stores the system's knowledge, employing an ontology to model the workspace. This ontology

defines relevant entities, their properties, relationships, and a set of rules that govern decision-making.

- Inference Engine (IE): The IE processes input data by applying rules from the KB. It evaluates the conditions and executes the corresponding rules to produce appropriate output responses.
- System Interface (SI): The SI acts as the user interface, allowing non-expert users or external applications to interact with the system. Through the SI, users can input new data and access the output of the system.

The structure of the AI module is highly versatile. The KB can be developed in a scalable manner, adapting to the specific physical system under consideration. The integration of a real machine tool and a cobot ensures a thorough analysis of human-machine interaction while keeping the number of rules at a manageable level. In more complex scenarios, such as real industrial environments, the KB can be expanded accordingly with additional rules, while the hardware components will scale in parallel to accommodate the increased system complexity. The development process of the AI module begins with creating a formal representation of the workspace using an ontology. This ontology organizes the system's relevant entities as classes, along with their instances, properties, and relationships. For example, Table 1 illustrates how the lathe is represented within the ontology, demonstrating how the workspace is modeled to enable accurate and context-aware decision-making. This structured approach ensures that the AI module effectively integrates domain expertise, regulatory requirements, and real-time data to support advanced safety functions. By leveraging this methodology, the system can deliver robust and precise responses tailored to the operational context.

After defining the ontology, the complete set of rules is developed, with each rule structured as: IF condition THEN action. These rules are based on a thorough risk analysis, which identifies and describes all reasonably foreseeable hazardous situations. Each rule condition specifies a unique combination of property values that could lead to such situations. For instance, when the lathe is powered on ($power = "ON"$) and operating in automatic mode ($operative_mode = "AUTO"$), the door guard must remain closed. To enforce this safety requirement, the system includes a rule to verify the condition. If the condition is not met, a "door guard tampering" warning is triggered. The rule is expressed as follows:

```
IF lathe.power == "ON"
AND lathe.operative_mode == "AUTO"
AND door_guard.state == "OPEN"
THEN door_guard.is_tampered = True
```

During runtime, the AI module receives real-time data from the IoT sensors. The Inference Engine (IE) evaluates the incoming data against the predefined rules, activating those whose conditions are met. Based on the evaluation, the module responds by either:

- Sending a stop signal to the production systems to ensure safety.
- Notifying operators through an alarm to prompt immediate attention via a mobile device, such as smartphone.

The redundancy of the system significantly enhances the overall reliability of the apparatus. Specifically, the integration of multiple detection mechanisms results in a more robust hypothesis testing framework. By combining CV and RFID, it is possible to simultaneously improve both specificity and sensitivity, thereby reducing erroneous state diagnoses. For instance, in the context of door tampering detection (Table 2), the safe state of the door guard (state A) is verified through both CV and RFID, leading to a diagnosis (B: door guard closed) with a joint probability $P(A \cap B)$, which is higher than the probability achieved by either system alone (enhanced specificity). Similarly, an unsafe state (A-, door guard not safe) is detected by both CV and RFID, yielding a corresponding diagnosis (B-, door guard not closed) with a joint probability $P(A - \cap B -)$, thus improving overall sensitivity. Since an incorrect diagnosis would require both systems to fail simultaneously, the combined system exhibits greater sensitivity compared to individual detection methods. This approach ensures proactive detection and mitigation of hazardous situations, enhancing the overall safety and reliability of the system.

4. Description of the use cases

This section presents a series of operational scenarios designed to test the functionality of the prototype system, with the primary goal of identifying potential hazards and preventing incidents. The UCs were developed by a multidisciplinary team of expert technicians, leveraging diverse expertise from both academic and industrial backgrounds. The team comprises specialists in machinery safety and practical applications of AI systems, ensuring a comprehensive approach to safety analysis. Thanks to the risk assessment, the UCs were conceived to simulate potential hazardous situations, enabling the evaluation of the system's effectiveness in maintaining operational safety using the instructions for use for the definitions of the standard operations and reasonably foreseeable misuses. For each scenario, the team identified a specific unsafe

Table 2
Hypothesis testing outcomes for door guard safety monitoring.

CASE	DOOR STATE (A)	DIAGNOSIS (B)	JOINT PROBABILITY	INTERPRETATION
1	Safe (A)	Door guard closed (B)	$P(A \cap B)$	True positive
2	Safe (A)	Door guard not closed (B')	$P(A \cap B')$	False negative
3	Unsafe (A')	Door guard closed (B)	$P(A' \cap B)$	False positive
4	Unsafe (A')	Door guard not closed (B')	$P(A' \cap B')$	True negative

Table 1
Representation of the lathe according to the defined ontology.

ENTITY	CLASSES	PROPERTIES	PROPERTY VALUES	INSTANCES
Lathe	Lathe	has guards power operative_mode	List of Guard instances "ON", "OFF"	lathe
	Guard	presence position is_removed	"MANUAL", "AUTO", "SETUP", "SERVICE" "DETECTED", "UNDETECTED" "MOUNTED", "UNMOUNTED"	—
	UnmovableGuard	is_subclass_of	True, False Guard	cart_guard tailstock_guard
	MovableGuard	is_subclass_of state is_tampered	Guard "OPEN", "CLOSED" True, False	door_guard spindle_guard

condition (e.g., tampering with safety guards, exceeding the allowable number of operators, cyber interference with the cobot) along with the associated hazards. The conditions identified in this study represent typical safety-critical situations that may occur in real industrial manufacturing environments, where operator safety is a major concern. For example, the tampering of safety guards on lathe machines is a well-documented and recurrent issue, often associated with severe operator injuries. However, it is important to emphasize that the defined UCs are currently tested under controlled laboratory conditions. As a result, several industrial boundary conditions commonly encountered in real-world environments, such as poor housekeeping, insufficient or variable lighting, and mechanical vibrations are not explicitly considered at this stage. These factors may affect the reliability and predictivity of the proposed system. For instance, low visibility levels or the presence of dirt and debris may degrade the performance of the Computer Vision subsystem. Considering the operative frequency of the tag RFID, electromagnetic interferences are not considered a possible issue of the application. In real industrial deployments, such conditions of reduced visibility could reduce detection reliability or increase the likelihood of missed or delayed safety responses. Consequently, while the proposed approach demonstrates functional feasibility under controlled conditions, careful system tuning, robustness enhancement, and environmental adaptation will be required for out-of-laboratory implementation.

The scenarios also outline how the system's components - machine control systems, CV, and RFID technology - can provide signals to monitor and supervise the prototype system. The overarching idea behind these scenarios is to effectively integrate the system's components to create a more robust and high-performing safety solution. The testing phase plays a critical role in observing system behavior under real-world production conditions, verifying its reliability and robustness, and ultimately enhancing workplace safety for operators.

As described in Section 3, the prototype system comprises a CNC lathe and a collaborative robot working in tandem to automate the turning process of various workpieces stored in a bin near the workstation. Notably, real-time monitoring of the distance between workers and machines throughout the operational phase is essential to maintain safety standards.

4.1. Defeating guards

The first scenario addresses the potential defeat of interlocks on interlocked movable guards or the removal of fixed guards. Such situations expose the machine's moving components during manual operations, posing significant safety risks. For further information on the defeating of guards, refer to EN ISO 14119 (ISO, 2013). The presence and status of the guards are monitored using both a CV system and an RFID system. The CV system evaluates the status (open/closed) of the

door, the tool, and the chuck by detecting specially designed ArUco markers affixed to the guards and the machine frame (Fig. 6). This system verifies both the presence and positioning of the ArUco markers. When the movable guard is open, the QR code on the machine frame should be obscured and no longer visible to the CV system. Simultaneously, the ArUco markers on the guards change position, that the CV system detects and interprets accordingly. In parallel, the RFID system monitors the presence and orientation of RFID tags attached to the movable guards (Fig. 7). These tags are equipped with a single-axis accelerometer, which detects changes in acceleration along the z-axis (for instance, normally equal to $-g$ when the door guard is closed). Variations in the acceleration value indicate changes in the tag's orientation, allowing the system to detect and track movements of the movable guard effectively. Throughout this process, the production system identifies and indicates the current operational phase. The monitoring process is conducted whenever the work phase involves the opening or closing of a guard and continues iteratively until a definitive response is obtained. In the event of detected tampering, the system immediately issues a stop command (or emergency stop) to the assembly's safety control system. This response is considered a SF provided by the manufacturer. Similarly, if the CV or RFID systems detect the absence of one or more ArUco markers or RFID tags, the system identifies a "guard removed" status, triggering the same immediate stop response.

From the rules' viewpoint, here is an example of how these situations can be coded:

```
IF door_guard.sensor_state == 'CLOSED'
AND (cv_detected_data.door_guard_state == 'OPEN'
OR rfid_detected_data.door_guard_state == 'OPEN')
THEN door_guard_state_conflict.has_occurred = True
```

This example illustrates a scenario where, if the door guard switch indicates the door is closed (`door_guard.sensor_state == 'CLOSED'`) but the AI system detects the door as open through either the CV system (`cv_detected_data.door_guard_state == 'OPEN'`) or the RFID system (`rfid_detected_data.door_guard_state == 'OPEN'`) a door guard state conflict is flagged (`door_guard_state_conflict.has_occurred = True`).

4.2. Counting the number of people

This scenario addresses the presence of unauthorized operators during critical operations, such as setup and maintenance. The involvement of individuals lacking the necessary training and the presence of a number of workers higher than the maximum allowed pose a significant safety risk to themselves and their colleagues. Consequently, developing a system capable of identifying and counting operators in the working area represents a valuable advancement in

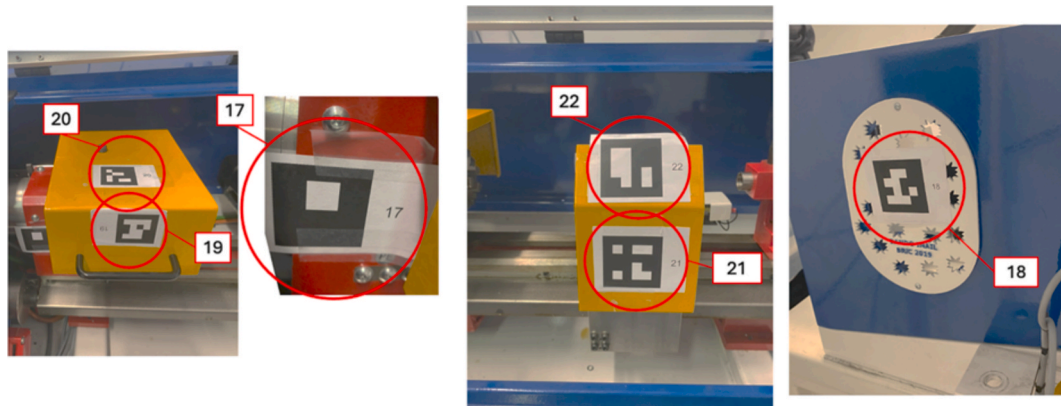


Fig. 6. Movable and fixed guards with ArUco markers.

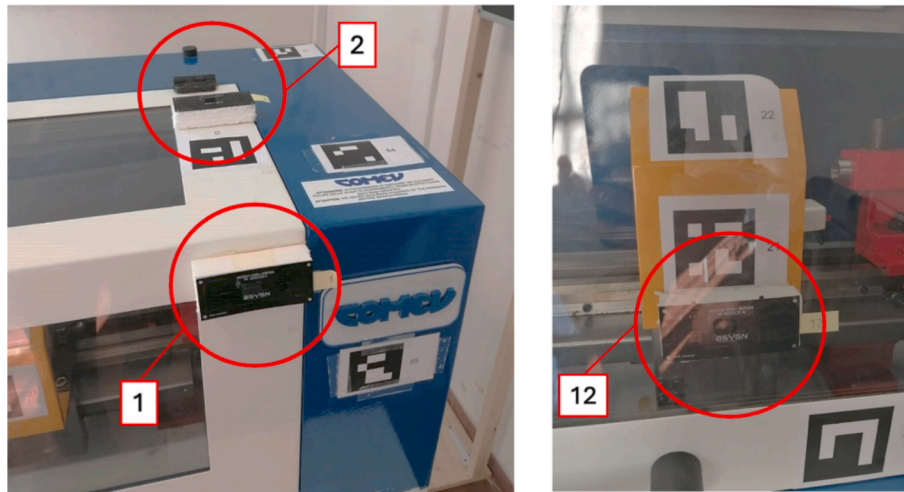


Fig. 7. RFID tags on the movable guards.

operational safety. To this purpose, the authors propose the following solution:

- The CV system performs real-time counting of individuals in the working area using specialized objects and people detection algorithms. Near the lathe, potentially hazardous areas have been identified where the presence of workers during various operational phases could pose a risk. The zones are defined using ArUco markers (Fig. 8). Each area is spatially delineated by placing ArUco markers at its four corners. For detecting individuals, advanced AI-based object detection models are employed, offering state-of-the-art reliability and performance in dynamic environments. This integrated approach combines the accuracy of person detection systems with the versatility and efficiency of ArUco marker technology for area tracking, providing a scalable and customizable solution for workplace safety monitoring. The system's person-counting algorithm operates as follows:
 - o For every person detected within the camera's field of view, a bounding box is generated to indicate their position.
 - o The algorithm calculates the percentage of overlap between the bounding box and predefined areas (e.g., Area A and Area B).
 - o For each identified person, the system sends an informational string to the AI module, including the person identifier and the

percentage of overlap between the bounding box and the defined area.

- Simultaneously, the RFID system tracks both the number and identity of operators using RFID tags embedded in their coveralls or other PPE. The tags are strategically placed to ensure that at least one tag can be detected by the antennas, regardless of the operator's orientation. It is important to note that the RFID-UHF system operates within a narrow broadcast range, which limits its ability to provide precise localization using fixed antennas. Instead, parameters such as the reading rate from the antennas are more effective for accurately counting personnel.
- This combined approach allows for the verification of both the number of people present and their authorization levels. The maximum allowable number of personnel and the required authorizations are determined by the ongoing operational phase, as communicated to the AI module by the machine control system. For this scenario, the monitoring frequency should operate within a time interval on the order of seconds. Additionally, discrepancies between the counts provided by the CV and RFID systems may indicate the presence of unauthorized individuals in the working area or the failure of an operator to wear the appropriate PPE as specified in the machine user manual. In such cases, a warning signal will be sent to the HSE supervisor and the line manager. It is important to note that operator identification in this system does not involve the collection of biometric or other personally identifiable data, thereby ensuring privacy compliance.

Here is an example of how a rule for detecting operator risk could be defined:

```
IF (lathe.operative_mode == 'MANUAL' OR lathe.operative_mode == 'SETUP')
AND (NOT(cobot.is_enabled) OR cobot.enable_mode == 'MANUAL')
AND (((area_1.number_operators + area3.number_operators) > 1)
OR (area_2.number_operators > 0))
THEN operators_risk_detection.has_occurred = True
```

This example specifies that if the lathe is in either manual or setup mode ($\text{lathe.operative_mode} == \text{'MANUAL'}$ OR $\text{lathe.operative_mode} == \text{'SETUP'}$), and the cobot is either disabled ($\text{NOT cobot.is_enabled}$) or in manual mode ($\text{cobot.enable_mode} == \text{'MANUAL'}$), while there are either more than one operator in areas 1 and 3 combined ($\text{area_1.number_operators} + \text{area_3.number_operators} > 1$) or at least one operator in area 2 ($\text{area_2.number_operators} > 0$), then a risk detection for operators is triggered ($\text{operators_risk_detection.has_occurred} =$



Fig. 8. Field of view of the Raspberry Pi 5 camera with delimitation ArUco marker.

True).

4.3. Manumission of the cobot

This scenario addresses the risk of cobot tampering with a corresponding unauthorized movement due to cyberattacks, accidental errors, or intentional interference. According to the MR, it is mandatory to safeguard the cobot in such situations. Tampering with the program of the cobot could result in unexpected movements, potentially causing collisions with the operator. To ensure safety, both the software and operational data must be protected, especially when a change in conditions compromises the compliance of the quasi-machine or machine. The supervision control system ensures that the cobot's movements are authorized either by the operator, using the proper control console, or by the lathe's control system. Specifically, it verifies that the cobot behaves as programmed, including activation only during specified times. To achieve this, the following control activities are implemented. First, the machine control system communicates the current operational phase, determining whether the cobot should be active. The CV system tracks ArUco markers located on the joints (Fig. 9), verifying that they align with the expected predefined positions. The RFID system, equipped with accelerometer-integrated tags, tracks the current positions and movements of the cobot. Similar to the tampering guard use case, these tags detect joint movements by measuring variations in acceleration along their sensing axis. Additionally, the phase difference between the tags can be utilized to enhance the identification of the joints' positions.

These systems work in tandem to ensure the actions of the cobot are authorized, whether initiated by the operator through a control console or by the machine itself. An AI module integrates signals from the CV system, RFID system, and the machine control system. By analyzing and comparing these inputs, the AI module detects anomalies or discrepancies. If inconsistencies arise, such as a mismatch between at least two signals, the system triggers a "Tampered Cobot" status. Upon detecting this status, the system initiates an immediate stop command or an emergency stop, depending on the risk level. This response, as in Section 3.1, constitutes a SF provided by the manufacturer.

Here is an example illustrating the logic and the consequentiality of the rules:

```
IF ai_module.receive.cv_detected_data.cobot_state == 'REST'
AND   ai_module.receive.rfid_detected_data.cobot_state ==
'NOT_REST'
THEN cobot_position_state_conflict.has_occurred = True
```

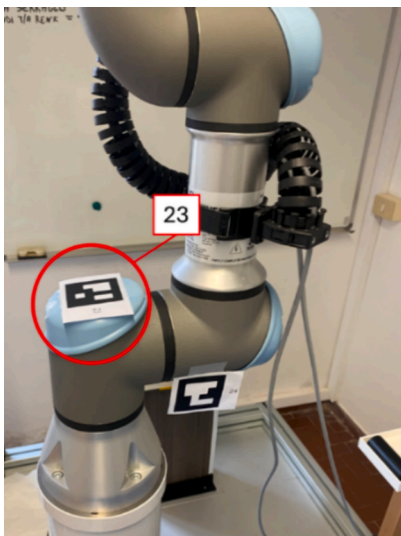


Fig. 9. ArUco markers on the cobot's joints.

In this example, the first rule checks for conflicting data regarding the cobot's state. If the AI module receives information from the Computer Vision system indicating the cobot is in a rest state (`ai_module.receive.cv_detected_data.cobot_state == 'REST'`), but simultaneously receives data from the RFID system indicating it is not in a rest state (`ai_module.receive.rfid_detected_data.cobot_state == 'NOT_REST'`), the system flags a cobot position state conflict (`cobot_position_state_conflict.has_occurred = True`). This conflict triggers a safety response:

```
IF cobot_position_state_conflict.has_occurred
THEN ai_safety.write.robot_emergency_ai_signal.digital_value = 0
```

In this second rule, the detected cobot position state conflict (`cobot_position_state_conflict.has_occurred`) leads directly to a safety measure. The AI safety module issues an emergency signal to the robot, setting the digital value of `robot_emergency_ai_signal` to 0. Please note that since the stop signals sent by the AI system to the production systems are safety signals, their digital value under normal (non-hazardous) conditions is set to 1. Therefore, in the event of risk detection, the rule that is triggered must "drop" this signal to a digital value of 0. For this reason, in the "risk" rules presented here, the action corresponds to resetting the stop signal's digital value to 0.

4.4. Obscuring cameras

This scenario investigates the potential obscuration of the camera-based vision system, specifically through the placement of a static image in the captured area. This scenario is critical, as tampering with the CV system could enable operators to engage in improper behavior, potentially resulting in hazardous situations. To determine whether the CV system has been obscured, the supervision system performs the following tasks:

- **Static Image Detection:** The CV system evaluates consecutive image acquisitions to identify whether the captured image remains unchanged over time, indicating possible tampering.
- **Movement and Presence Verification:** The RFID system monitors the number of individuals in the work area using tags embedded in their coveralls, while the CV system assesses the captured image for operator movement or stillness.

The monitoring process operates at a frequency on the order of seconds to ensure timely detection. If the AI system receives signals from the RFID system indicating the presence of individuals but no corresponding movement or activity from the CV system, it detects a tampering action on the CV system. Upon detecting such tampering, the system immediately issues a machine stop command or an emergency stop to the control system, ensuring safety and compliance. Here is an example highlighting the logic and consequentiality of the rules:

```
IF (frontal_camera.saw_frozen_image
OR right_camera.saw_frozen_image
OR ceiling_camera.saw_frozen_image)
AND orientation_antenna.has_detected_some_movement
THEN cv_blackout.has_occurred = True
```

This rule checks for a potential blackout in the Computer Vision system. If any of the cameras (frontal, right, or ceiling) detect a frozen image (`frontal_camera.saw_frozen_image`, `right_camera.saw_frozen_image`, or `ceiling_camera.saw_frozen_image`), and the orientation antenna simultaneously detects movement (`orientation_antenna.has_detected_some_movement`), a Computer Vision blackout is flagged (`cv_blackout.has_occurred = True`). This detected blackout triggers an immediate safety response:

```
IF cv_blackout.has_occurred
```

```
THEN ai_safety.writes.general_emergency_ai_signal.digital_value = 0
```

In this second rule, the occurrence of a Computer Vision blackout (`cv_blackout.has_occurred`) leads directly to the activation of a safety protocol. The AI safety module issues a general emergency signal by setting the `general_emergency_ai_signal` digital value to 0.

4.5. Monitoring of components presence and position

This scenario analyzes the presence and positioning of critical components within the working area, specifically focusing on a container used for collecting waste chips produced during the machinery (Fig. 10). The absence or incorrect positioning of the container poses potential risks to the operator, including exposure to moving machine parts and the accumulation of sharp chips on the floor (poor housekeeping), which can lead to safety hazards. The assessment of the presence and position of the container is performed using a combination of CV and RFID systems. The CV system detects the presence and position of the container by recognizing specially designed ArUco markers affixed to its lateral faces (red circles), using advanced image processing algorithms. The RFID system complements this by verifying the container's presence and position through RFID tags embedded on the container itself (green circle). The proposed solution involves the following steps:

- **CV Monitoring:** The CV system continuously monitors the ArUco markers placed on the container to confirm its presence and correct positioning.
- **RFID Detection:** The RFID system cross-checks the container's presence and position by detecting the embedded RFID tags.

This monitoring process is independent of the operational phase of the machine and should occur at intervals of approximately one minute. If the system identifies the absence or incorrect positioning of the chip container, it triggers an alarm signal to the operator and halts further processing steps until the issue is resolved. Here is an example emphasizing the logic and consequentiality of the rule:

```
IF lathe.power == 'ON'
AND NOT (container.is_in_correct_position)
THEN container_removal.has_occurred = True
```

This rule monitors the position of the container relative to the lathe's operational state. If the lathe is powered on (`lathe.power == 'ON'`) but the container is not in its correct position (`NOT(container.is_in_correct_position)`), a container removal event is flagged (`container_removal.has_occurred = True`).



Fig. 10. Monitoring presence and position of the chip container.

5. Risk analysis

Defining specific risk scenarios is a critical first step, as these scenarios guide the selection and placement of sensors. This study focuses on common hazards encountered in real work environments. For each identified UC, one or more sensors are assigned to monitor specific parameters, with monitoring frequency determined by the technological capabilities of the sensors. Equally important is defining the type of response to be triggered when a hazardous situation is detected. Fig. 11 illustrates the contributions of each IoT sensor for the specified UCs with reference to the MANUAL operational state.

Given that the Numerical Control (NC) of the lathe provides four distinct operational modes - Manual, Automatic, Setup, and Service - each with unique safety conditions, additional matrices have been developed to accommodate these variations. These matrices specify the required sensors and responses for each operational mode. Defining UCs and their corresponding technologies also facilitated the identification of a set of I/O signals required for communication between the AI system and the other components of the prototype, including the CV system, RFID system, and production control system. The primary objective is to develop a prototype that operates as a conventional machine assembly during standard tasks while providing enhanced safety features when activated by the responsible operator. This approach ensures compliance with the MD guidelines and prevents unsafe operations, such as engaging in the robot's collaborative mode, when the monitoring and supervision system is inactive. To activate the system, the responsible operator must use the mode selector on the control console to send an "Enable AI" signal to the AI module. The module then verifies the statuses of the IoT sensors to confirm whether the monitoring and supervision task can begin. Upon successful verification, the AI module sends a return signal, labeled "AI-Ready," to the PLC module of the lathe terminal, using Normally Closed (N.C.) logic. If this return signal is interrupted or if a discrepancy is detected between the dual-channel outputs, the production system halts in accordance with the monitoring and supervision system's safety protocols. If this return signal is interrupted or if a discrepancy is detected between the two outputs, the production system will cease following the system's safety instructions. In the event of a signal fault, the safety of the lathe PLC promptly detects the issue, preventing potential safety hazards. This layered approach ensures robust communication and safe operation, leveraging IoT sensors and AI for enhanced safety and reliability. As illustrated in Fig. 12, the electric circuits of the supervisor and production systems are isolated using relays to prevent electrical issues, such as short circuits, from impacting both systems. The safety signal is transmitted from the lathe to the robot's safety terminal, highlighted in yellow, ensuring seamless integration of safety protocols across all components.

For details on stop function categories, refer to EN 60204-1 (IEC, 2018). When a hazard involves only the robot, the lathe remains unaffected, allowing its work cycle to continue uninterrupted. In such cases, a Category 2 stop is especially useful, as it halts the robot while preserving the current phase of the lathe's cycle. Conversely, a Category 0 stop, which removes power entirely, provides a higher level of safety but may prevent resuming the process from the last phase, potentially resulting in the loss of the workpiece - a critical consideration in lengthy or high-value processes. The safety signal ultimately reaches the robot terminal to trigger the appropriate safety response. According to the user manual, Universal Robots e-Series robots include integrated safety functions and I/Os designed for connection with other machines and additional protective devices. These features comply with ISO 13849, achieving Performance Level d (PLd) with Category 3 architecture. The robot supports the following types of stops, as defined by the EN 60204-1 standard:

- Emergency Stop → Stop Category 1.
- Safety Stop → Stop Category 2.
- Limit Violation → Stop Category 0.

		OPERATIVE MODE: MANUAL									
		CV		RFID			LATHE				
USE CASES	SUBCASE	PRESENCE	POSITION	PRESENCE	POSITION	ORIENTATION	SPINDLE GUARD	DOOR GUARD	MAN. EN.	AUT. EN.	POWER ON
MOBILE GUARD TAMPERING	SPINDLE GUARD TAMPERING										
	DOOR GUARD TAMPERING										
	SPINDLE GUARD REMOVAL										
	DOOR GUARD REMOVAL										
OPERATOR'S COUNT	PRESENCE										
INFORMATIC TAMPERING ROBOT	UNEXPECTED MOVEMENT										
CV TAMPERING	BLACKOUT										
FIXED GUARD TAMPERING	CART GUARD										
	TAILSTOCK GUARD										
TAMPERING RFID	TAMPERING										
PRESENCE DPI	PRESENCE										
	POSITION										
OPERATOR'S CONDITION	POSITION										
	ORIENTATION										
OBJECT POSITION	WORKPIECES STORAGE										
AUXILIARY SYSTEM PRESENCE	CHIPS STORAGE										

Fig. 11. Use Cases for the manual operational state of the lathe.

- Anomaly Detection → Stop Category 0.

6. Conclusions

In January 2027, the new EU Machinery Regulation will come into effect, marking a transformative shift in machinery safety by introducing provisions for the integration of AI-based technologies to ensure safety functions. This regulation also addresses the critical dynamics of human-machine coexistence within collaborative working environments.

This paper presents an innovative AI-driven monitoring and supervision system designed to manage operational safety in production processes. The proposed system integrates three key components: an RFID-based tracking system, a CV system, and a machine control system. The RFID system identifies the position- and, in certain cases, the orientation and acceleration - of RFID tags embedded within operators' PPE or affixed to key surfaces in the production environment. Communication is facilitated through a network of antennas strategically positioned around the machine assembly. The CV system provides continuous monitoring of the working area via cameras equipped with advanced object and person recognition algorithms. The processing time

of computer vision depends strongly on the current type of hardware devices which could be optimized during a scale-up phase improving its prompt capability of anomaly detection. The machine control system, governed by Safety-Related Software, processes data from wearable devices and cameras to manage safety-critical situations. Central to this architecture is an AI module implemented as a Rule-Based Expert System. The module utilizes an ontology to store domain knowledge and applies inference rules to analyze real-time data. This multilayered monitoring approach enhances system redundancy, significantly reducing the likelihood of interpretation errors - particularly false negatives - that could otherwise jeopardize operator safety. To validate this AI-based supervision system, an experimental setup was developed comprising a multimodal CNC lathe and a collaborative robot, which works alongside an operator to execute workpiece movements and turning operations. The system's performance will be evaluated across diverse scenarios thanks to risk assessment of the standard operations and possible reasonably foreseeable misuses from the instructions for use, assessing responsiveness to dynamic conditions and its capability to promptly implement corrective actions. The structure of the smart supervision system here proposed is very effective in the automatic detection of risky operative situations only if they are well anticipated so

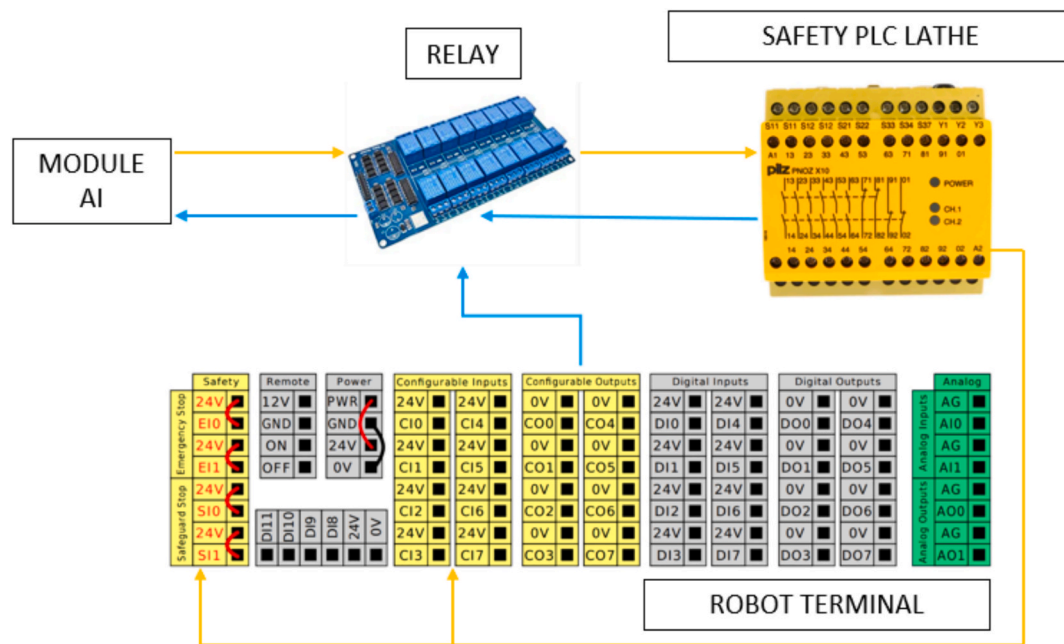


Fig. 12. Electric connection between lathe and robot.

that the specific disposition of markers and tags on the machine and the relative sequence of rules are implemented and activated.

It is important to stress that the integration of the smart supervision here proposed does not affect the usability of the production system limiting the false stop commands. So, false stop commands, which could degrade the user trust in the system due to the loss of productivity (i.e. loss of usability) and might induce the user to defeat it, are practically negligible. Moreover, the supervision system has the intrinsic capability to detect possible defeating issues.

Building on these results, future research will extend the proposed methodological framework beyond stationary industrial equipment. In particular, the same AI-based supervision and safety architecture will be applied within a new research project, named MACHINE5.0, which focuses on mobile machinery operating in outdoor construction environments. In this context, AI and autonomous technologies will be deployed on moving vehicles, enabling the assessment of the proposed approach under significantly higher levels of environmental variability, uncertainty, and interaction complexity. This evolution toward real-world construction scenarios will represent a crucial step toward practical deployment, contributing to the advancement of intelligent, and regulation-compliant safety systems for next-generation industrial and construction machinery.

CRedit authorship contribution statement

Francesco Di Paco: Writing – original draft, Conceptualization. **Luca Burattini:** Investigation, Conceptualization. **Roberto Gabbrielli:** Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Luca Landi:** Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Francesco Marcelloni:** Supervision, Software, Methodology, Investigation, Conceptualization. **Leonardo Marrazzini:** . **Marco Palumbo:** Software, Data curation, Conceptualization. **Marco Pirozzi:** Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Roberto Gabbrielli reports financial support was provided by National

Institute for Insurance against Accidents at Work. Marco Pirozzi reports a relationship with National Institute for Insurance against Accidents at Work that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors wish to thank the Istituto Nazionale Assicurazione Infortuni sul Lavoro (INAIL – Italy, website www.inail.it) for its financial support of the project AISAFETY “Integrated smart system based on artificial intelligence for the management of operator safety in production processes” through the fund “Bando BRiC. Bando Ricerca in Collaborazione (BRiC – 2022) – Piano Attività di Ricerca 2022-2024”.

Data availability

No data was used for the research described in the article.

References

- Ayhan, B.U., Tokdemir, O.B., 2019. Safety assessment in megaprojects using artificial intelligence. *Saf. Sci.* 118, 273–287. <https://doi.org/10.1016/j.ssci.2019.05.027>.
- Barbosa, F.G.O., Mourao, G.L., Erazo, J.J.M., Ghedini, G.M., Valverde, J.A., 2025. Dynamic risk management in office workspaces: Real-time analysis of staircase safety using computer vision and AI. *Comput. Electr. Eng.* 122, 109902. <https://doi.org/10.1016/j.compeleceng.2024.109902>.
- Bostrom, A., Demuth, J.L., Wirz, C.D., Cains, M.G., Schumacher, A., Madlambayan, D., Bansal, A.S., Bearth, A., Chase, R., Crosman, K.M., Ebert-Uphoff, I., Gagne, D.J., Guikema, S., Hoffman, R., Johnson, B.B., Kümmler-Bonfanti, C., Lee, J.D., Lowe, A., McGovern, A., Williams, J.K., 2024. Trust and trustworthy artificial intelligence: A research agenda for AI in the environmental sciences. *Risk Anal.* 44 (6), 1498–1513. <https://doi.org/10.1111/risa.14245>.
- Buffi, A., Bernardini, F., Nepa, P., Marracci, M., Tellini, B., Di Donato, L., Pirozzi, M., Tomassini, L., Ferraro, A., 2021. RFID-Based Localization Enables a Smart System for Worker Safety. In: *2021 IEEE 6th International Forum on Research and Technology for Society and Industry (RTSI)*, pp. 513–518. <https://doi.org/10.1109/RTSI50628.2021.9597296>.
- Buyse, L., Habli, I., Vanoost, D., Pissort, D., 2025. Safe autonomous systems in a changing world: Operationalising dynamic safety cases. *Saf. Sci.* 191, 106965. <https://doi.org/10.1016/j.ssci.2025.106965>.
- Cagno, E., Accordini, D., Neri, A., Negri, E., Macchi, M., 2024. Digital solutions for workplace safety: An empirical study on their adoption in Italian metalworking SMEs. *Saf. Sci.* 177, 106598. <https://doi.org/10.1016/j.ssci.2024.106598>.

- Choi, B., Hwang, S., Lee, S., 2017. What drives construction workers' acceptance of wearable technologies in the workplace?: Indoor localization and wearable health devices for occupational safety and health. *Autom. Constr.* 84, 31–41. <https://doi.org/10.1016/j.autcon.2017.08.005>.
- Condry, M.W., 2023. Technologies of Industry 4.0 and Opportunities They Create for Product Innovation. *IEEE Eng. Manag. Rev.* 51 (4), 185–190. <https://doi.org/10.1109/EMR.2023.3315657>.
- European Parliament and of the Council. (2006). Directive—2006/42—Machinery Directive—EUR Lex. *Official Journal of the European Union*.
- European Parliament and of the Council. (2023). Regulation (EU) 2023/1230 of the European Parliament and of the Council. *Official Journal of the European Union*, 66.
- Garyaev, A., Garyaev, N., 2023. Integration of artificial intelligence and video surveillance technology to monitor construction equipment. *E3S Web of Conferences* 410, 04002. <https://doi.org/10.1051/e3sconf/202341004002>.
- Guo, B.H.W., Zou, Y., Fang, Y., Goh, Y.M., Zou, P.X.W., 2021. Computer vision technologies for safety science and management in construction: A critical review and future research directions. *Saf. Sci.* 135, 105130. <https://doi.org/10.1016/j.ssci.2020.105130>.
- Hwang, J., Lee, K., Ei Zan, M.M., Jang, M., Shin, D.H., 2023. Improved Discriminative Object Localization Algorithm for Safety Management of Indoor Construction. *Sensors* 23 (8), 3870. <https://doi.org/10.3390/s23083870>.
- IEC. (2018). *EN IEC 60204-1: Electrical equipment of machines—Part 1: General Requirements*.
- IEC. (2019). *IEC TS 62998-1:2019 Safety of machinery—Safety-related sensors used for the protection of persons*.
- ISO. (2010). *UNI EN ISO 12100:2010 Safety of machinery—General principles for design—Risk assessment and risk reduction*.
- Iso, 2012. —Safety of machinery—Risk assessment Part 2: Practical guidance and examples of methods. *ISO/TR 14121-2*, 2012.
- Iso, 2013. *Safety of machinery—Interlocking devices associated with guards—Principles for design and selection*. *ISO 14119*, 2013.
- ISO. (2015). *EN ISO 13849-1:2015—Safety of machinery—Safety-related parts of control systems Part 1: General principles for design*.
- ISO. (2024). *BS EN ISO 13855:2024-Safety of machinery-Positioning of safeguards with respect to the approach of the human body*.
- Kim, K., Kim, M., 2012. RFID-based location-sensing system for safety management. *Pers. Ubiquit. Comput.* 16 (3), 235–243. <https://doi.org/10.1007/s00779-011-0394-0>.
- Landi, L., Buffi, A., Marracci, M., Stecconi, A., Leone, P. D., Bernardini, F., & Didonato, L. (2021). Localization Systems for Safety Applications in Industrial Scenarios. *Proceedings of the 31st European Safety and Reliability Conference (ESREL 2021)*, 2413–2419. Doi: 10.3850/978-981-18-2016-8_261-cd.
- Le, T.-H., Lin, J.J., 2025. AR-enhanced near-miss incident alert system for onsite moving equipment. *Saf. Sci.* 191, 106912. <https://doi.org/10.1016/j.ssci.2025.106912>.
- Phatangare, S., Kate, S., Khandelwal, D., Khandetod, A., & Kharade, A. (2023). Real Time Human Activity Detection using YOLOv7. *2023 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, 1069–1076. Doi: 10.1109/I-SMAC58438.2023.10290168.
- Pishgar, M., Issa, S.F., Sietsema, M., Pratap, P., Darabi, H., 2021. REDECA: A Novel Framework to Review Artificial Intelligence and Its Applications in Occupational Safety and Health. *Int. J. Environ. Res. Public Health* 18 (13), 6705. <https://doi.org/10.3390/ijerph18136705>.
- Rashidi, A., Woon, G.L., Dasandara, M., Bazghaleh, M., Pasbakhsh, P., 2025. Smart personal protective equipment for intelligent construction safety monitoring. *Smart Sustainable Built Environ.* 14 (3), 835–858. <https://doi.org/10.1108/SASBE-10-2022-0224>.
- Reyes, M.A., King, G.W., Miller, G.G., 2014. Intelligent Machine Guard Monitoring: A Wireless System to Improve Miner Safety. *IEEE Ind. Appl. Mag.* 20 (2), 69–75. <https://doi.org/10.1109/MIAS.2013.2288400>.
- Ruz, M.L., Vázquez, F., Salas-Morera, L., Cubero-Atienza, A., 2012. Robotic testing of radio frequency devices designed for industrial safety. *Saf. Sci.* 50 (7), 1606–1617. <https://doi.org/10.1016/j.ssci.2012.03.018>.
- Stødle, K., Flage, R., Guikema, S., Aven, T., 2025. Artificial intelligence for risk analysis—A risk characterization perspective on advances, opportunities, and limitations. *Risk Anal.* 45 (4), 738–751. <https://doi.org/10.1111/risa.14307>.
- Tan, H., 2017. A brief history and technical review of the expert system research. *IOP Conf. Ser.: Mater. Sci. Eng.* 242, 012111. <https://doi.org/10.1088/1757-899X/242/1/012111>.
- Trivedi, P., Alqahtani, F.M., 2024. The advancement of Artificial Intelligence (AI) in Occupational Health and Safety (OHS) across high-risk industries. *Journal of Infrastructure, Policy and Development* 8 (10), 6889. <https://doi.org/10.24294/jipd.v8i10.6889>.
- Xu, Z., Huang, J., Huang, K., 2023. A novel computer vision-based approach for monitoring safety harness use in construction. *IET Image Proc.* 17 (4), 1071–1085. <https://doi.org/10.1049/ipr2.12696>.
- Yang, X., Yu, Y., Shirowzhan, S., Sepasgozar, S., Li, H., 2020. Automated PPE-Tool pair check system for construction safety using smart IoT. *Journal of Building Engineering* 32, 101721. <https://doi.org/10.1016/j.jobe.2020.101721>.
- You, M., Li, S., Li, D., Xu, S., 2021. Applications of artificial intelligence for coal mine gas risk assessment. *Saf. Sci.* 143, 105420. <https://doi.org/10.1016/j.ssci.2021.105420>.